Farmer App — Pakistan | Product Source of Truth & PRD

2026-10-02 Final Product Clarification — Identity, Consent & Merchant Credit Boundary

Identity and passive activity

- Reading, browsing, searching, viewing recommendations, weather, market information, crop information and similar passive/non-accountable activity remains available without forced signup where product/business rules allow.
- Identity is requested **at the point of action** when the system needs attributable ownership, communication, publication, persistent history, financial accountability or another authenticated write.
- Typical identity-gated actions include posting/replying, account-linked uploads, dealer/expert messaging, credit requests/acceptance, field-visit requests and other durable accountable actions.
- This does not change the Farmer App's guest-first model.

Shop Owner merchant-credit boundary

- The MVP credit feature is based on the existing Pakistan shop-owner/farmer commercial practice represented by this product.
- The **Shop Owner/Dealer is the commercial decision-maker and counterparty** for the credit terms in the MVP.
- The Farmer App facilitates the request, offer, acceptance and status workflow; it does not introduce a separate lending product at this stage.
- Shop Owner-defined terms shown through the product must be explicitly presented to the farmer, and the farmer must provide consent before

submitting the request and again to accept any changed counteroffer.

- The exact terms, applicable charges/conditions and business rules remain attributable to the Shop Owner/Dealer; the product must preserve the accepted terms/version as an audit record.
- Future scaling, broader financing models, regulated lending, additional financial products or automated underwriting are explicitly **future architecture/business work** and must not be invented in the MVP.

Privacy/consent

- Consent is required where the product collects/uses data for a purpose that requires user permission, including applicable privacy, AI/media and communication disclosures.
- Final legal wording/policy text can be reviewed separately; the product behavior must support explicit consent records, policy/terms versioning and auditable acceptance.

Provider/commercial approval — simplified meaning

- This is **not a product question for the owner**.
- It simply means: when we use an outside service such as weather data, the project must have the correct account/subscription/API credentials and permission to use that provider's data in a commercial app.
- For weather, the current implementation candidate is a paid commercial Weatherbit plan; its current pricing shows commercial-use licensing on paid plans. cite turn126132search0 turn126132search3
- For market rates, AMIS is the initial source candidate; production must use a permitted/approved access method rather than assuming that a public website automatically grants unrestricted API/data rights. AMIS is operated for agricultural price dissemination across Punjab markets. cite turn126132search9
- Architecture owns this dependency and should handle provider research/configuration; escalate only if an actual business purchase, contract, or legal approval is needed.

2026-10-02 Identity-on-Action + Flow Simplification Rules

Identity-on-action principle

The Farmer App should follow a **minimum-identity / action-gated identity** model:

- Farmers may enter and use the core utility without an account.
- Passive activity such as browsing, reading, viewing recommendations, weather, market information, crop information, and other non-identity-dependent experiences should not force signup.
- Low-risk local interactions may remain guest-capable where they do not require durable account attribution.
- Any action that creates, publishes, sends, owns, commits, or persistently changes user/business data requires authenticated identity at the point where that identity becomes necessary.
- Examples include posting, replying/commenting, uploading content that is associated with the farmer account, sending dealer/Expert messages, requesting credit, accepting a credit offer, requesting a field visit, saving persistent history, or other identity/accountability-dependent actions.
- The app should ask for identity **at the moment it becomes necessary**, not earlier just to increase signup friction.
- When a guest has already built useful farming context, safe context migration into the authenticated account should happen without unnecessary re-entry.
- Identity requirements must never be inferred from UI placement alone; the backend authorization and business operation determine whether identity is required.

Flow simplification / step-change rule

An implementation agent may simplify a flow from 3 steps to 2, combine screens, reorder non-dependent inputs, or remove genuinely redundant UI steps **without changing the product's underlying rules**.

A flow step may not be removed or skipped when it is the only place where the system:

- collects required business data;
- captures required consent/terms acceptance;
- confirms a state transition;
- establishes authorization/identity;
- performs required validation;
- records an audit decision;
- prevents an unsafe/duplicate action;
- preserves a required source-of-truth relationship.

If a required piece of information can be collected in another safe step, the flow may be compressed while preserving the same data, validation, permission, state, audit and outcome contracts.

Impact rule

Changing the **presentation/sequence** of a flow does not automatically change:

- agricultural knowledge;
- farmer/crop facts;
- behavioral signals;
- recommendations;
- dealer/business source of truth;
- credit rules;
- expert verification rules;
- notification truth;
- permissions/security.

Those systems change only when the modified flow changes the underlying data, event meaning, state transition, or business rule.

Before implementing a requested flow simplification, the agent must compare:

old flow → required inputs → state transitions → validations → permissions → data writes → source of truth → downstream consumers → analytics/notifications → E2E regression.

If all contracts remain equivalent, the change is a **UX/flow simplification**, not a product-rule change.

If any contract changes, the agent must surface the impact before implementation rather than silently redefining the product.

2026-10-02 Research Review + Implementation Blueprint — Architecture/Research Findings

Research conclusion

The Farmer App is implementable on the accepted stack (**React Native + Expo + TypeScript + Supabase/PostgreSQL + Edge Functions + OpenAI**) without introducing a second product architecture. The critical design is to keep each business/data domain's source of truth explicit and put external providers/knowledge sources behind adapters.

1. Agricultural knowledge system — recommended baseline

Use a curated evidence pipeline rather than unrestricted web/LLM knowledge:

Authoritative / trusted source registry → ingestion → normalization → agricultural entities → evidence records → expert review → approved knowledge → retrieval → AI interpretation

Strong source families verified during research include:

- Punjab Agriculture Department production technologies, farmer extension/training material, research institutes and pest-warning services.
- Pakistan Department of Plant Protection registered pesticide information and registration processes.
- CABI/PlantwisePlus crop-health and pest-management knowledge.
- FAO digital-agriculture/AI guidance emphasizing contextual, localized and responsibly governed agricultural knowledge.

Punjab Agriculture Department explicitly maintains approved production technologies, farmer extension programs and pest/crop-disease warning services. cite turn157665search3 turn157665search1 turn157665search2

DPP publishes a registered-pesticide list and describes the formal pesticide-registration process. cite turn992107search0 turn992107search2

CABI's PlantwisePlus Knowledge Bank provides country/global plant-health information for farmers and advisors.
cite turn157665search9 turn157665search8

FAO's current digital-agriculture roadmap emphasizes contextual adaptation, accountability, security and data stewardship, while its farmer-AI work describes localized datasets rather than unrestricted internet knowledge.
cite turn992107search9 turn992107search8

2. Launch agricultural coverage — recommended implementation

Do **not** restrict which crops a farmer can add.

Instead use coverage tiers internally:

- **Validated / high-confidence:** strong recommendations enabled.
- **Partially validated:** informational answers only, with confidence/source disclosure and escalation where necessary.
- **Unsupported:** acknowledge limited verified knowledge; do not manufacture treatment/action.

For the first Punjab knowledge build, prioritize major/high-value crop domains represented in current Punjab agricultural programs and research, including **wheat, cotton, rice, maize and sugarcane**, then expand into fruits/vegetables and other locally important crops. This is a knowledge-priority order, not a farmer eligibility list. cite turn157665search0 turn157665search14 turn157665search17

3. Pesticide/product safety boundary

The system must never jump directly from:

crop problem → dealer product ID.

Required chain:

crop/problem/symptom → approved treatment/active ingredient knowledge → applicable crop/use evidence + regulatory/label evidence → canonical product → live dealer listing

A dealer product being in stock is not evidence that it is agronomically appropriate.

Registered-pesticide information should be treated as evidence for product eligibility where applicable. cite turn992107search0 turn992107search13

4. Market / mandi data

AMIS Punjab is a strong initial source because Punjab's official technology board describes AMIS as disseminating agricultural commodity prices from **135 markets across Punjab** and maintaining historical reporting. Punjab's portal also lists the AMIS application as providing daily commodity prices by city. cite turn545971search4 turn545971search0

Implementation:

AMIS source/access method → ingestion adapter → normalize → validate/dedupe → historical observations → farmer UI + AI + recommendation layer

Important rule:

- Do not assume a public API exists merely because an AMIS app/site exists.
- Confirm the permitted/approved machine-readable access method before production ingestion.
- Preserve source + observation date/time + market + unit + min/max/FQP where provided.
- Missing market/crop observations remain missing; never invent a price.

5. Weather

The weather domain should remain provider-abstracted.

Research found:

- **Open-Meteo free API is explicitly non-commercial**, so it should not be treated as the production source for a commercial Farmer App without the

appropriate paid/commercial arrangement. cite turn992107search3

- Weatherbit offers commercial-use plans and an agriculture-weather offering, making it a practical provider candidate for evaluation. cite turn569431search2
- WeatherAPI provides current + forecast APIs with paid commercial plans and can be evaluated as a fallback. cite turn569431search8

Recommended candidate: evaluate **Weatherbit** first, with the internal provider adapter preserved so the product does not depend on one vendor.

Weather UI and AI weather answers must consume the same normalized weather service and source timestamp.

6. Language + voice

Text support can remain:

Urdu default + Punjabi + Saraiki + Sindhi + Pashto.

Voice must be quality-gated rather than promised blindly.

Current Google Cloud Speech documentation lists speech recognition for Urdu (Pakistan), Pashto, Punjabi and Sindhi variants, but **Saraiki is not listed** in the supported Speech-to-Text languages found during research. Its Text-to-Speech catalog also has language/voice coverage that is not equivalent across these languages. cite turn306136search0 turn154729search1 turn154729search3

Implementation rule:

- language selection is independent of voice availability;
- all five languages can be supported in text;
- voice input/output is enabled only where the selected provider has acceptable quality;
- Saraiki voice requires an additional validated provider/solution before being promised in production;
- run real-device farmer-language QA before enabling each language's voice path.

7. Expert verification

The user-facing expert profile should show evidence, not merely a badge:

- identity/basic profile;
- qualification/degree;
- certification(s);
- issuing authority;
- supporting document;
- agricultural specialization;
- years/type of experience;
- relevant location/language coverage;
- platform tenure/contribution history;
- verification status + date.

HEC currently operates an online degree-attestation workflow that includes document upload, university/HEI verification and e-attestation, so HEC-attested evidence can be used as a stronger verification input when relevant. cite turn465444search0 turn465444search61

Architecture must still support manual issuer verification because not every agricultural credential will have an API/public verification endpoint.

8. Credit / legal boundary

The finalized credit UX is a **merchant trade-credit workflow**, not a general-purpose lending platform:

Farmer request → Shop Owner review → optional single counteroffer → Farmer accept/reject → Shop Owner final accept/reject → transaction commit

For safety and legal separation, the Farmer App should not itself become the creditor in the initial model, set hidden fees/penalties, underwrite automatically, or hold/disburse loan principal. The Shop Owner remains the commercial decision-maker/business counterparty unless a later legal/business decision explicitly changes that model.

Pakistan's SECP currently maintains a regulated digital-lending framework and a current whitelist for apps operated by licensed lending NBFCs. That makes it

important to obtain legal confirmation on the exact boundary between merchant trade credit and regulated lending before expanding the credit model. cite turn992107search1 turn992107search5

9. Terms & Conditions architecture

Recommended:

- Shop Owner can define merchant-specific credit terms allowed by the product.
- Farmer sees the relevant terms **before submitting the request**.
- Store a version/snapshot of the exact terms accepted.
- If a counteroffer changes price/repayment condition, the resulting offer should show the changed values clearly and use the appropriate terms version.
- The Farmer App should provide a standard platform disclaimer/notice while not inventing the shop owner's legal clauses.

10. Community moderation

Final operating model remains:

Report → Admin/Moderation queue → human review → explicit action → audit

Recommended additions:

- severity/category on the moderation case;
- evidence/context snapshot;
- action reason;
- actor + timestamp;
- separate notification behavior for reporter vs content author;
- AI-assisted detection/prioritization only.

Never make “report” itself equal “permanent delete.”

11. Data, privacy and retention

Current public official sources show Pakistan's comprehensive personal-data framework remains an area requiring legal verification: MoITT's legislation page

lists the 2023 Personal Data Protection Bill as a draft, while older/other parliamentary material reflects an evolving legislative process. Therefore, do not hard-code a legal assumption into the product; design to strong privacy-by-default principles and confirm the governing law before production policy is finalized. cite turn201938search1 turn201938search0

Implementation:

- minimum necessary collection;
- explicit consent where needed;
- purpose-specific access;
- user/account deletion workflow;
- raw-media time limits;
- automatic retention jobs;
- no unrestricted session replay for sensitive credit/AI content.

Supabase Cron can schedule retention/deletion jobs directly from Postgres/Edge Functions. cite turn561253search7

12. Media architecture

Use:

capture → client compression → upload → processing state → optimized display media + analysis-quality copy where needed → retention job

Supabase Storage currently supports image resizing/optimization and private signed transformed URLs; this fits profile/community media well. cite turn561253search0 turn561253search9

For videos, keep a separate transcoding/processing path rather than assuming image transformations cover video.

13. Location + dealer search

Use PostGIS in Supabase:

- farmer/default search point stored as geographic point;
- maximum radius = 100 km;

- geospatial filter + distance sorting;
- live dealer eligibility/business status;
- product live/availability check;
- rank with proximity + availability + verification + operational/order-performance signals.

Supabase documents PostGIS specifically for efficient geo queries and indexed nearest-neighbor/location filtering. cite turn720178search0

14. Analytics / session intelligence

Recommended candidate: **PostHog** for product analytics + AI event analytics, with technical error monitoring and backend logs separated from product analytics.

PostHog currently supports product analytics, funnels, retention, paths, session replay, React Native, and React Native error tracking. cite turn569431search10 turn312506search0 turn312506search9

However:

- do not send raw AI conversation/media/credit payloads into analytics;
- use structured events such as intent, crop, recommendation ID, outcome, screen and workflow state;
- sensitive session replay should stay disabled unless explicitly designed and masked.

15. Offline launch recommendation

Do **not** attempt “fully offline Farmer App” for MVP.

Recommended:

- cache profile/farmer context;
- cache active crop state;
- cache recent recommendations;
- cache last-known weather/market values with freshness labels;
- allow safe local drafts for farmer notes/community composition where useful;

- queue only low-risk context updates;
- require live connectivity for AI diagnosis, dealer inventory, messaging delivery, credit actions, and authoritative transaction changes.

This gives resilience on weak connectivity without creating a second offline source of truth.

16. Source-of-truth architecture

The project should use these canonical domains:

Farmer identity

→ Farmer App Auth/Profile

Farming context

→ Crop Instance / Crop Lifecycle

Agricultural knowledge

→ Approved Knowledge + Evidence

Weather

→ Weather Provider Adapter

Market

→ Market Observation Adapter

Dealer

→ Dealer/Shop Owner system

Inventory/product listing

→ Dealer system

Business transaction / credit status

→ Dealer system

Recommendation

→ Canonical Recommendation/Insight service

Community content

→ Farmer App Community service

Expert verification

→ Verification record + Admin/Moderation decision

Moderation

→ Moderation case + action audit

Notifications

→ Delivery layer referencing canonical recommendation/event

17. End-to-end implementation chain

Every feature should be implemented and verified as:

Final UI → component contract → data model → source of truth → state machine → backend/RPC/Edge Function → authorization → persistence → notifications → analytics → E2E → regression evidence

No implementation agent should create a parallel source of truth simply because a screen needs data.

18. Recommended initial architecture diagram

Mobile App

→ Auth / Local Cache / UI / i18n / Media

Supabase

→ Postgres + RLS + PostGIS + Storage + Realtime + Cron

→ Edge Functions / RPCs

Domain services

→ Farmer Context

→ Crop Lifecycle

→ Knowledge/Evidence

→ Recommendation Engine

→ Weather Adapter

→ Market Adapter

→ Dealer Integration

→ Credit Workflow

→ Community/Moderation

→ Expert Verification

→ Notifications

→ Analytics/Observability

External systems

→ OpenAI

→ Weather provider

→ AMIS/Punjab source

→ DPP/approved knowledge sources

→ Dealer business system

→ Optional speech provider(s)

Research verdict

The product architecture is coherent and implementable.

The remaining work is no longer broad product discovery. It is:

1. a small set of owner/business decisions;
2. provider/legal confirmations;
3. turning the locked UI into exact screen/data/state contracts;
4. implementation + evidence-based QA.

2026-10-02 Finalized Expert Verification + Community Moderation + Credit Workflow

Expert verification - finalized operating model

- A person may apply to become a **Verified Expert**, or the platform may identify a candidate and invite them to apply.
- Verification evidence should capture, where applicable:

    - identity/basic profile information;
    - professional qualification(s) and certification(s);
    - issuing institution/body;
    - certificate/document evidence;
    - agricultural specialization/areas of expertise;
    - years/type of relevant experience;
    - relevant location/coverage;
    - platform tenure and contribution history;
    - quality/evidence history of community advice and farmer interactions.
- Community participation may strengthen reputation but **must not by itself create professional expert verification**.
- Expert verification is distinct from:
    - Verified Dealer;
    - Community Contributor/Reputation.
- Initial verification is manual.
- **Final approval, suspension, and removal authority belongs to the internal Admin/Moderation function.**
- Verification records should be auditable and should retain the evidence/reviewer decision used to grant or remove the badge.
- Certificate/document expiry or re-verification behavior remains an operational policy to define later; the data model must support expiry/review without redesigning the expert profile.

Community moderation - finalized control model

- Any formal community report creates a moderation case and surfaces it to the internal **Admin/Moderation** workflow for review.
- The moderation system should highlight the reported content/reason and relevant context so the operator can make the decision.

- The operator may take an appropriate action such as:
  - no action / dismiss;
  - warn;
  - restrict or temporarily hide where supported;
  - remove content;
  - escalate a safety-sensitive case.
- Permanent removal must be an explicit moderation decision, not an automatic consequence of every report.
- Moderation decisions and reasons should be recorded for audit/history.
- User notifications about moderation outcomes should be controlled by the moderation decision and product policy, not blindly sent before review.
- AI may help detect/prioritize potentially harmful content, but it does not independently award expert status or make irreversible moderation decisions.

Credit - finalized farmer/shop-owner workflow

The Farmer App does **not** merge Farmer and Dealer/Shop Owner identities. Credit is a workflow-level integration between two independent systems.

Initial request

1. Farmer selects a product/order context and requests credit.
2. Before submission, the farmer is shown the applicable shop-owner terms and conditions.
3. The farmer must explicitly confirm that they have reviewed/accepted the displayed terms version.
4. The request is submitted to the Shop Owner/Dealer system.
5. The system prevents duplicate concurrent credit requests for the same underlying order/transaction context.

Shop Owner decision

- The Shop Owner can:
  - accept the farmer's requested terms; or
  - create **one counteroffer** by changing allowed commercial terms such as price and/or repayment condition/date.
- A counteroffer is sent back to the farmer with a clear record of what changed.
- The Farmer can only **Accept** or **Reject** that counteroffer.
- The Farmer cannot create a second counter-counteroffer.
- The Shop Owner cannot repeatedly counteroffer within the same request. The workflow is intentionally **not a negotiation loop**.

Final approval

- If the farmer rejects the counteroffer, the request ends without creating the credit transaction.
- If the farmer accepts the counteroffer, the request moves to **Awaiting Shop Owner Final Approval**.
- The Shop Owner then performs the final **Accept** or **Reject** action.
- Only the Shop Owner's final acceptance creates/commits the approved credit transaction in the integrated business workflow.
- If the Shop Owner rejects at final approval, the request ends without creating the approved credit transaction.
- The implementation must use an explicit state machine and atomic business action so an accepted offer cannot be committed twice.

Canonical record / audit

The credit request must preserve:

- original farmer request terms;
- any single shop-owner counteroffer;
- farmer accept/reject decision;

- final shop-owner accept/reject decision;
- terms/version shown to the farmer;
- acceptance timestamps;
- actor/source for each decision;
- resulting business transaction reference when finally accepted.

Credit state model

Requested → Shop Owner Reviewing → Counteroffered → Awaiting Farmer Decision → Awaiting Shop Owner Final Approval → Accepted / Rejected

Where no counteroffer is made:

Requested → Shop Owner Reviewing → Awaiting Shop Owner Final Approval → Accepted / Rejected

Expired/cancelled states may be added later if business policy requires them; the state model must remain extensible.

Legal/commercial boundary

- No separate lending product is being introduced in this MVP.
- The **Shop Owner/Dealer defines the merchant credit terms** for the transaction.
- The Farmer must explicitly review and consent to the displayed Shop Owner terms before submitting the credit request.
- If a Shop Owner sends a counteroffer with changed terms, the changed terms must be shown clearly and the Farmer must explicitly accept or reject that offer.
- The product records the accepted terms/version and decision history for auditability.
- Final legal wording/policy review remains an operational/legal dependency, not a reason to reopen the product workflow.

- Future scaling into broader financing/lending products is a separate future business + architecture phase.

2026-10-02 User Module + Conversation Retention Rules

User Profile module

Guest state

- Profile entry remains available before signup, but the guest must not see account-only controls as if an account exists.
- Guest can see/edit applicable local preferences and farming context that the product allows without an account.
- Guest profile should provide a clear path to create/login to an account.
- **Delete Account** is not shown for a guest because there is no authenticated account to delete.

Authenticated state

The current final Profile design should remain visually unchanged. Its underlying module should support the existing profile information and settings:

- profile image
- farmer name
- mobile number
- location
- farmer/crop context shown by the design
- My Dealers
- Farm details/context where applicable
- Language
- Notifications

- Help & Support
- Logout
- Delete Account

Account identity fields are distinct from farming context. A farmer can update editable profile/context fields without changing historical transaction records owned by the dealer system.

Profile/location behavior

- The farmer's default location is a reusable context value used by Home/weather/personalization and as the default starting point for dealer search.
- Changing the temporary dealer-search location must not silently change the farmer's default profile location.
- Location remains editable after onboarding.

AI question/history experience

- User-visible AI conversations/history are a **future-capability** unless already present in the locked final UI.
- Do not build an infinite archive by default.
- The structured farmer context and useful recommendation history are retained separately from raw media.
- Search/AI/community behavior should create compact structured signals for personalization rather than requiring permanent retention of every raw event payload.

Recommended initial retention policy

These are architecture recommendations to be validated against privacy/legal requirements:

- Structured confirmed farming context, crop lifecycle state, important recommendations and key outcomes: retain while relevant to the farmer account, subject to account deletion/privacy policy.

- User-visible text conversation history: retain only as long as product value justifies it; because text is storage-light, it does not need aggressive deletion solely for storage reasons.
- Behavioral personalization signals: strongest relevance window around **90 days**, then decay rather than remaining equally influential forever; older history can remain as low-weight historical context or aggregate topic statistics.
- Raw voice recordings: target **7 days** maximum unless a specific case requires longer retention.
- Uploaded AI-analysis images/videos: target **30 days** maximum after the structured result is generated, unless explicitly required for an active case.
- Technical traces/debug payloads: target **14–30 days**, with only necessary aggregates retained longer.
- Failed/temporary uploads should be cleaned automatically.
- Retention jobs must be automated; deletion must not depend on manual cleanup.

Personalization memory model

raw interaction → temporary processing data → structured signal → relevance/recency scoring → personalization → retention/decay

The system should retain the **signal** longer than the raw artifact whenever possible.

Exact profile labels

The final UI remains the visual source of truth. Any label whose exact text is not recoverable from the design evidence must not be invented by engineering; preserve the final design label when implementing.

2026-10-02 Farmer → Shop Owner Field Visit / Crop Inspection Request

- A farmer can send a **field visit / crop inspection request** from a shop owner's/dealer's profile.

- The purpose is for the farmer to request an agricultural representative/field person associated with that shop/dealer to visit the farmer's field, inspect the crop, identify the reported issue, and provide practical guidance.
- This is a distinct service workflow; it is **not normal chat and not credit**.
- The farmer initiates the request and provides the minimum information needed to understand the visit, with richer details captured later if required.
- The shop owner reviews the request and can **Accept** or **Reject** it.
- On acceptance, the shop owner can confirm that a field person will visit. Detailed scheduling (date/time, assigned person, visit confirmation, rescheduling, completion, visit notes) is intentionally a later workflow extension.
- Initial lifecycle: **Pending** → **Accepted / Rejected**. The model must be extensible to add **Scheduled** → **In Progress** → **Completed / Cancelled** later without rewriting the original request.
- The farmer must always be able to see the current request status and next action.
- Any field-visit findings or advice captured later should become a structured crop/issue record, with source and timestamp, so the result can feed the farmer's crop history and future recommendations rather than remaining only inside chat.
- Professional field-inspection/advisory services are a real agricultural operating model in Punjab/Pakistan; government extension services include crop pest scouting/advisory, while commercial agricultural organizations also provide field visits and technical field support. cite turn565614search13 turn565614search6

2026-10-02 Confirmed Product Decisions — Geography, Languages, Dealer Radius & Media

Geography

- Initial operational geography: **all Punjab**.

- South Punjab, including Saraiki-speaking pilot users, is an important initial validation population within Punjab.
- The product remains architected for Pakistan-wide expansion; Punjab is the initial operational/validation boundary.

Language

- **Urdu is the default language** for the Farmer App and AI assistant.
- Initial regional-language support must include **Punjabi, Saraiki, Sindhi, and Pashto**, with the ability for the farmer to manually select their preferred language.
- Location can suggest a regional language, but must not silently change the farmer's selected language.
- AI text and voice responses should follow the selected language where quality is sufficiently validated.

Nearby dealer search

- Maximum supported dealer-search radius: **100 km**.
- Default search location is the farmer's configured location, but the farmer may change the search location for a particular search.
- The radius is a circular geospatial boundary around the selected search location.

Media compression

- Images and videos must be compressed before upload to reduce mobile data usage, storage and upload time, while preserving sufficient quality for the intended use.
- Do not use one fixed compression level for every media type.
- **Normal social/community/profile media:** adaptive resize + compression optimized for fast loading; a WhatsApp-like reduction in file size is the target experience rather than preserving original camera files by default.

- **AI crop/diagnosis media:** use a higher-quality analysis copy so compression does not remove important visual symptoms. The system may create separate optimized display and analysis representations from the same capture.
- Compression should happen client-side before upload where practical, with server-side validation/transcoding as a safety net.
- Preserve the original only when there is an explicit product/business reason; otherwise keep optimized representations.
- Video should be constrained by practical maximum duration/resolution/file size and transcoded to an efficient mobile format; exact limits are an implementation proposal to be validated against device/network conditions.
- Media processing status must be represented explicitly so the user knows whether upload, compression, processing, or AI analysis is still in progress.

Privacy / raw-media retention

- Useful structured farming history, recommendations and confirmed context may persist according to the product's retention policy.
- Raw voice recordings, uploaded images/videos and technical traces must be retained only for a **defined limited period**, then automatically deleted unless an explicit retention exception exists.
- Proposed architecture defaults: short-lived raw technical traces and temporary media, with only the structured result/history retained longer. Exact retention periods require final privacy/legal confirmation.

Architectural principle

Optimize media for the user's network first, but never compress away information required for safety or diagnosis. AI-analysis quality takes precedence over aggressive compression when the media is being used for crop/problem identification.

2026-10-02 Personalization + Market Intelligence Update

Behavioral personalization loop

- The platform should continuously learn from farmer interactions to make future content/recommendations more relevant.
- Signals may come from: searches, AI conversations, scan topics, crop questions, community participation, product searches, dealer searches, viewed recommendations, saved content, dismissed recommendations, and explicitly confirmed crop/context changes.
- The goal is **farmer usefulness**, not maximizing time spent in the app.
- Relevance should be calculated using a combination of crop association, explicit farmer context, recent intent, repeated interest, recency, current crop stage, location relevance, and recommendation urgency.
- Behavioral signals must decay over time; an old question should not permanently dominate personalization.
- A single interaction should not be treated as a strong fact. Repeated or explicit evidence can increase a signal's confidence.
- Personalization must remain explainable at the system level: every surfaced recommendation should be traceable to the farmer/crop context and the evidence that made it relevant.
- The system should not use engagement optimization to push irrelevant content simply because the farmer is likely to click it.

Personalization flow

Farmer activity → structured signal → signal strength/recency → crop/context association → candidate content/recommendations → relevance + safety + freshness checks → Home/notification/search suggestions.

Notification relationship

- Personalization may identify that a topic is becoming more relevant, but notification delivery still requires a separate importance/actionability/frequency check.
- A relevant interest signal does not automatically create a notification.

- The notification must reference a real underlying content/recommendation/event object.

Market / mandi price system

- Market rates are a separate data domain and must not be invented or generated by AI.
- Initial source candidate: AMIS Punjab. The AMIS portal exposes agricultural commodity wholesale/auction prices by market and date, including min, max and FQP values for reported commodities. cite turn630423search0 turn630423search1
- The backend should use a market-data adapter/ingestion layer so the Farmer App consumes a normalized internal format rather than depending directly on AMIS page structure.
- Minimum normalized record: commodity/crop, market, date/time of observation, unit, min price, max price, FQP/representative price where provided, source, source URL/reference, ingestion timestamp, freshness/status.
- Store historical observations; do not overwrite yesterday's rate with today's rate.
- Missing market observations must remain missing; never fill a missing mandi price with an invented value.
- Show source + observation date/time wherever rates are presented so the farmer knows whether the number is current or historical.
- Market queries from AI should call the same market data service used by the Market UI.
- Future providers can be added behind the same adapter if AMIS coverage, availability, licensing, or reliability is insufficient.
- Any price trend/prediction is a derived analytical layer and must be clearly separated from the observed market rate.

Market insight loop

Observed market data → normalization/quality checks → farmer location/crop relevance → market-rate UI → optional derived trend/alert → notification only when actionable and sufficiently fresh.

New non-negotiable

Observed data, inferred signal, agricultural recommendation, personalized relevance, and transactional/business record are separate objects. A personalization signal can influence what we surface; it cannot rewrite the source-of-truth data.

2026-10-02 Final Product/System Review — Core Architecture Decisions

Purpose: Convert the owner's latest product decisions into implementation-safe system rules. This section supersedes earlier interpretations where they conflict.

1. Audience vs implementation coverage

- The Farmer App is intended for farmers broadly; farm size/acreage is **not** an onboarding eligibility gate.
- This does **not** mean every crop/disease recommendation must be enabled on day one. The product may support any farmer while only giving high-confidence, validated agronomic recommendations for knowledge domains that have been verified.
- When reliable knowledge is unavailable, the system must say so rather than manufacture a recommendation.

2. Crop-first context; no mandatory Farm → Field hierarchy for MVP

- Do not force a farmer-facing Farm → Field setup during onboarding.
- The farmer can directly select one or more crops and provide their relevant timing/season information.
- Backend should represent each active crop as a **Crop Instance / Crop Lifecycle** so each crop has independent state.

- An optional area/field/farm abstraction may be added later if real product needs justify it.
- Crop lifecycle should track, where available: crop, cycle/season, planting/start date, estimated/current growth stage, activities, problems, inputs/treatments, harvest, yield/outcome, and relevant signals.
- Crop stage must never be treated as certain merely because elapsed time suggests a stage. Source/confirmation must be tracked.

3. Agricultural state: confirmed facts vs inferred signals

Create two distinct classes of farmer/crop knowledge:

- **Confirmed facts:** explicitly entered or confirmed by the farmer, or reliably recorded by an authoritative business/data source.
- **Inferred signals:** derived from behavior, questions, AI conversation, image analysis, timing, or other indirect evidence.

Inferred signals may personalize what the app asks/shows, but must not silently overwrite a high-impact canonical crop fact. Where a recommendation depends on uncertain stage/condition, the system should request confirmation.

Example:

Planting date → estimated stage = inference.

Farmer says “my cotton is flowering” → confirmed stage after confirmation.

AI sees possible flowering in image → candidate signal, not canonical fact until confirmed.

4. Crop lifecycle and personalization

- Multiple active crops are supported simultaneously.
- Recommendations/signals are evaluated per crop instance.
- A valid signal for Crop A must not suppress a valid signal for Crop B.
- Global anti-spam/frequency rules still apply.
- A **Featured/Primary Crop** may be selected explicitly by the farmer and may also be suggested by the system.

- System suggestion should use evidence such as recent activity, repeated queries, recency, and any explicitly provided importance/area information where available.
- Because acreage is not mandatory, the system must not pretend it knows which crop has the highest financial stake unless the farmer has provided evidence for that.

5. Agricultural AI boundary

The AI assistant is an **agriculture-domain assistant**, not a general-purpose chatbot.

Allowed AI knowledge/actions should be limited to the approved agricultural domain, including:

- crop/farming questions
- crop problems/pests/diseases
- cultivation and farming activities
- agricultural weather questions
- crop/season planning
- agriculture-related market/input questions
- the farmer's own farming context and history.

General non-agricultural questions (politics, sports, entertainment, unrelated general knowledge, etc.) are out of domain.

Product/account facts may be answered through deterministic product services rather than general AI knowledge.

Off-domain behavior:

- clearly state that the assistant is focused on agriculture;
- do not answer unrelated topics from general model knowledge;
- do not use general internet content as an unrestricted answer source.

6. AI architecture recommendation

Use a **knowledge-grounded agricultural decision-support architecture**, not a raw LLM answer path.

Core loop:

User/context → intent/domain gate → structured context extraction → authoritative/approved data + agricultural knowledge → deterministic rules/tools where required → AI interpretation/explanation → confidence/safety check → structured response.

Current global agricultural implementations support this pattern: FAO highlights localized, context-specific agricultural datasets and validated protocols rather than treating general internet knowledge as sufficient, while Plantix exposes structured diagnosis output with confidence and treatment information. cite turn715232search0 turn715232search7 turn715232search13

- AI may explain and interact.
- Canonical weather, inventory, price, dealer verification, transaction, credit status and other business facts must come from system sources.
- AI must never invent availability, price, eligibility, verification, or regulatory status.

7. Recommendation / Decision system

Create one canonical **Recommendation / Insight object** rather than recalculating the same recommendation independently on each screen.

Minimum conceptual fields:

- recommendation ID
- farmer/crop context
- type/problem
- priority
- recommendation
- evidence/inputs used
- source/knowledge version
- confidence

- valid-from / valid-until
- created/updated timestamps
- status: active / viewed / acted-on / dismissed / expired
- safety/escalation state.

The same object should power Home, notifications, crop details, recommendation history and relevant follow-up flows.

8. Weather

- Raw/current/forecast weather comes from the weather provider, not LLM memory.
- Agriculture-specific interpretation can be produced by the recommendation layer using weather + crop context.
- AI may answer farmer questions such as today's weather, tomorrow's rain possibility or week's forecast by retrieving current provider data.
- Weather-dependent recommendations must use the same provider data/source timestamps used by the visible weather experience.

9. Search

Use **intent-aware search**, not a generic undifferentiated search box.

Possible intents: agricultural information | weather | crop problem | product | dealer | community.

The system should route the query to the correct source. A weather query uses current weather data; a product query uses live dealer listings; a crop-disease question uses agricultural knowledge/diagnosis services.

10. Product recommendation and dealer inventory integration

- The dealer/shop-owner system remains the source of truth for dealer inventory/business data.
- Dealer/shop owner can mark a product/listing as **live for Farmer App** from the existing inventory system.

- Farmer-side product availability is a synchronized read model of the dealer system, not a second inventory truth.

Farmer-side fields can include:

- canonical product identity
- variant
- dealer
- price
- stock/availability
- dealer verification/operating status
- live/published status.

No delivery/pickup capability is part of the current design.

11. Critical product-matching requirement

Agricultural knowledge must not map a disease/pest directly to an arbitrary dealer inventory ID.

Use:

Crop/problem → approved treatment/product knowledge → canonical product → live dealer listings → location/radius/availability → farmer result.

A canonical product identity is required so multiple dealers can list the same real-world product safely.

For regulated pesticides, product knowledge should include registration/label evidence where applicable. Pakistan's Department of Plant Protection maintains a registered pesticide list and regulates pesticide registration/import under the Agricultural Pesticide Act/rules.
cite turn829182search0 turn829182search3 turn829182search16

The system must never recommend a chemical solely because a dealer happens to have it in stock.

12. Dealer discovery/ranking

The owner has chosen order volume, proximity and availability as ranking signals.

Recommended ranking model:

1. Eligible dealers within selected radius.
2. Dealer is active/operational and verified where applicable.
3. Required product is live and available.
4. Distance/proximity.
5. Historical order/transaction performance and responsiveness.

Order volume should be **one signal, not the only signal**, otherwise new verified dealers can be systematically buried.

Nearby is not automatically equivalent to recommended.

13. Dealer verification vs expert verification

Do not use community contribution alone as proof of professional expertise.

Recommended distinction:

- **Verified Expert:** credentials/identity checked through a defined verification process.
- **Community Contributor / Reputation:** earned through helpful participation and validated community feedback.
- **Verified Dealer:** business identity/status checked through the dealer verification process.

Contribution may strengthen reputation, but must not by itself create a professional-expert verification badge.

14. Dealer integration / farmer identity boundary

- Farmer account and dealer-side customer account remain independent.
- Do not globally merge the identities.
- Do not automatically synchronize every dealer transaction into a farmer account merely because phone numbers happen to match.
- When a farmer explicitly enters a dealer workflow through the Farmer App, the integration should carry a stable Farmer App interaction/customer reference

so the dealer-side transaction can be attributed safely.

- Farmer-facing purchase/credit history can then be shown as a synchronized read model while the dealer system remains the business source of truth.

15. Credit workflow — current business direction

- Dealer/shop owner decides credit eligibility.
- Farmer creates a credit request for an order/product.
- The request enters dealer review.
- Dealer may accept or modify/counter the commercial terms, including price and repayment dates/conditions.
- The updated offer returns to the farmer.
- Farmer can accept or reject.
- The farmer cannot create duplicate concurrent requests for the same order.
- Final agreement/acceptance must be distinct from dealer-side preliminary approval.
- After fulfilment, repayment/outstanding state is synchronized from the dealer system.
- When valid payment is recorded by the dealer system, Farmer App reflects the updated paid/outstanding status.
- Farmer-facing history should show what was bought, from which dealer, when, amount/price and current repayment status as permitted by the business/legal model.
- Real credit remains gated by final business/legal/payment decisions; do not invent financial rules in the app.

16. Farmer behavior / question signals

Capture useful structured signals such as:

- search intent/topic
- crop association

- scan/problem topic
- recommendation viewed
- recommendation acted on/dismissed
- repeated questions
- community topics engaged with
- dealer/product interests
- confirmed context changes.

Signals should have a source and timestamp. Behavioral evidence must not automatically become a confirmed agronomic fact.

This creates the personalization loop:

Farmer interaction → signal → context/personalization → relevant information/recommendation.

17. Notification synchronization

- The notification engine must consume the same canonical recommendation/event objects used by Home.
- Never implement separate notification business logic that invents a different version of the recommendation.
- A notification should reference the underlying recommendation/event ID so the app can open the exact same source state.
- When a recommendation becomes invalid/expired/acted-on, the corresponding notification state must update accordingly.

18. What is still genuinely open

Owner/business decisions still needed:

- pilot geography/initial district (this is a **pilot boundary**, not the definition of the overall farmer audience)
- initial high-confidence crop/disease coverage for launch, even if the app accepts any farmer

- initial supported languages
- maximum nearby-dealer search radius
- expert verification operational process
- community moderation/escalation policy
- exact weather/market data providers and commercial/legal approval
- credit legal/business model and payment rails
- privacy/data-retention/consent policy for farmer and AI interaction data.

Technical implementation of these items should be proposed by architecture; the owner should only decide business/product choices.

19. MVP principle

Do not build the whole agriculture universe before proving the core loop.

The implementation should first make this chain real and trustworthy:

Farmer context → current crop situation → understand problem → useful guidance → relevant live product/dealer → optional dealer interaction/credit → visible history.

Supporting services such as weather, search, notifications and community should connect to that same context/recommendation system rather than becoming isolated features.

20. Final system rule

Known fact ≠ inferred signal ≠ recommendation ≠ transaction record.

Each has its own source, confidence/authority, lifecycle and ownership.

This separation is mandatory for recommendation safety, synchronization and future scale.

2026-10-02 Product/System Verification Update — Onboarding, Identity, Location & Multi-Crop

Status: Accepted from owner discussion; must be reflected in implementation.

Onboarding / Account

- The Farmer App must be usable without account creation.
- Installation/onboarding must not force email, phone, OTP, or account creation before the farmer can enter the core farming experience.
- A farmer can optionally create/login to an account.
- Farming context can be collected before account creation so the Home experience can already be personalized.
- Guest and authenticated users should use the same underlying farming-context and recommendation logic; account status changes persistence/identity-required capabilities, not the core intelligence.
- When a guest later creates an account, their useful local farming context should be preserved/migrated where technically safe, rather than forcing re-entry.

Dealer-system boundary

- Farmer identity/account and dealer-side customer records are separate systems.
- Farmer signup/account creation must **not** query, match, link to, or autofill from the dealer-side customer database.
- The Farmer App owns the farmer profile.
- Dealer interaction/integration begins only when the farmer uses a dealer-related capability such as dealer messaging, product/order/credit workflows.
- Dealer-side customer information remains owned by the dealer system. Do not duplicate it into the Farmer App merely to support messaging/business workflows.
- Where an integration is required, use an explicit interface/service boundary and pass only the data required for that interaction.

Location

- During onboarding, request location permission so the app can establish a useful default location.

- If permission is denied/unavailable, allow the farmer to manually select an area.
- Until location is available, location-dependent UI should have a clear unknown/not-defined state rather than inventing a location.
- The farmer has a **default farming location** used for personalization/weather/context.
- Nearby-dealer search uses that default location initially but may use a different temporary search location without silently changing the farmer's default location.
- Dealer search radius must be configurable; maximum radius remains an open product decision.

Crops / personalization

- A farmer can have multiple active crops simultaneously.
- Each active crop is represented as its own crop instance/lifecycle with its own season/cycle, planting information where available, growth stage, activities, problems/signals, and recommendations.
- Recommendations and notifications must evaluate active crops independently. A recommendation for one crop must not suppress a valid recommendation for another crop.
- The system must still enforce global anti-spam/frequency rules so multiple relevant crop signals do not become excessive notification noise.
- Observed user behavior (searches, questions, scans, viewed information, etc.) may be stored as personalization/interest signals, but behavior must not automatically be treated as confirmed agricultural fact.

Architectural principle

Farmer context → crop lifecycle(s) → external/observed data → deterministic agricultural/business rules → AI interpretation where appropriate → canonical recommendation/signal → Home / notifications / relevant surfaces.

The UI does not invent recommendations, and separate surfaces must not independently calculate contradictory versions of the same recommendation.

1. Product Thesis

The Farmer App is a farmer-first daily utility for Pakistan.

Its primary job is not to sell agricultural products. Its primary job is to help a farmer make better decisions about crops, weather, pests/disease, farming activities, money, and trusted local support.

The commercial layer is intentionally secondary: when a farmer needs an input, the platform can connect that need to verified nearby agricultural shops and their real inventory.

Core relationship:

Farmer problem → understand/diagnose → explain what to do → connect to trusted local help/input → optionally transact → keep the farmer informed through the crop journey.

The app should earn usage through usefulness first. Commerce is a consequence of solving the farmer's problem, not the reason the farmer is forced into the app.

2. Product Vision

Build a trusted agricultural companion for Pakistani farmers that helps them answer:

1. What is happening to my crop?
2. What should I do now?
3. Is the timing/weather right?
4. What are other farmers saying?
5. What does a qualified expert say?
6. Where can I get the required input from a trusted local shop?
7. If I need credit, what exactly are both parties agreeing to?
8. What has happened with my crop, purchases, credit, and decisions over time?

3. Primary User

Initial target:

- Small and medium Pakistani farmers.
- Particularly farmers who already buy seeds, pesticides, fertilizer, and other inputs from local agricultural retailers.
- Low-to-moderate digital literacy.
- May use Urdu and/or regional languages.
- May use shared household devices.
- May have unreliable connectivity.
- Often operates outdoors and under strong sunlight.
- Needs quick, practical answers rather than long educational material.
- Trusts known people and local relationships heavily.

This is a working persona, not a claim that every Pakistani farmer behaves the same way. Validation should refine it.

4. Core User Problems

Crop problem recognition

A farmer notices disease, pests, discoloration, growth problems, weather damage, or another abnormality but may not know what it is.

Need: quickly understand the likely problem and the next action.

Timing

The farmer may know what to do but not when to do it.

Examples include irrigation, spraying, weather-sensitive activities, seasonal activities, and crop-risk timing.

Need: convert information into actionable timing.

Trustworthy inputs

A farmer needs to know whether a product and seller can be trusted.

Need: verified shops, transparent product information, real availability, and clear provenance/trust signals.

Local knowledge

Farmers already learn from other farmers, dealers, family, and experts.

Need: a structured place where farmer experience and qualified advice can coexist without pretending they have the same authority.

Money and credit

Credit can be important for input purchases. The farmer and dealer need clarity about what was requested, approved, supplied, owed, and when repayment is expected.

Need: a transparent, mutually visible credit process rather than an invisible or informal side agreement.

Information overload

A farmer does not need a generic encyclopedia.

Need: relevant information for their crop, location, season, and current situation.

5. Product Principles

1. Farmer-first, commerce-second.
2. Action over information.
3. Local relevance over generic content.
4. Trust must be visible, not assumed.
5. Expert advice and farmer opinion must be clearly distinguishable.
6. Never hide financial commitments.
7. Every important state must have a clear status and next action.
8. Design for low attention, low connectivity, outdoor use, and shared devices.

9. AI must communicate uncertainty instead of presenting guesses as facts.
10. Every major feature must have a clear reason to exist.
11. The system must preserve history when business state changes.
12. Do not create duplicate logic when a shared platform service already exists.

6. Product Architecture — Conceptual

The Farmer App is a farmer-facing client over shared platform capabilities.

The existing agricultural shop/dealer application remains the operational source for dealer inventory, customer relationships, credit records, and relevant business workflows.

The farmer app should consume shared services/data rather than creating a parallel dealer/customer truth.

Conceptual domains:

- Farmer identity/profile
- Farm/crop profile
- Crop lifecycle
- Weather/location
- Agricultural advisory/content
- AI crop assistant
- Community
- Expert responses
- Dealer discovery
- Product/inventory discovery
- Dealer communication
- Credit
- Orders/fulfilment
- Notifications

- History/audit
- Trust and moderation

Exact backend architecture is not yet finalized and must be decided deliberately before implementation.

7. Navigation / Information Architecture — Working Model

Working bottom navigation:

1. Home
2. Scan / AI
3. Community
4. Messages
5. Shop / Orders or another validated commercial destination

Profile lives behind the avatar/top-right account entry rather than occupying a primary bottom-nav slot.

This is a working information architecture, not a frozen decision. Validate against the actual screens and journeys before implementation.

8. Home

Purpose: answer within seconds:

- Is there anything important for my crop today?
- What should I do?
- Is there a weather-related action?
- Do I need help?
- What is happening with my money/orders?

Candidate modules:

- Today's crop/action insight

- Weather + actionable farming recommendation
- Scan crop/problem CTA
- Crop status
- Upcoming farming activity/reminder
- Market/crop rates
- Credit summary when relevant
- Dealer/order status when relevant
- Important alerts
- Recent community activity

Design rule:

Do not show every available data point.

Rank information by:

Urgency → impact → relevance → frequency → actionability.

9. AI Crop Assistant / Scan

Purpose: allow a farmer to photograph a crop/problem and receive understandable guidance.

Flow:

1. Farmer opens Scan.
2. Captures or uploads a crop image.
3. System collects necessary context where required: crop, location, growth stage, symptoms, and optional farmer description.
4. AI analyzes the case.
5. Result communicates likely issue, confidence/uncertainty, why the system thinks this, immediate next steps, and what information is still needed.
6. Farmer can ask follow-up questions.

7. If appropriate, the system can surface relevant products and verified nearby dealers.
8. If confidence is low or the case is high-risk, route to human/expert review rather than pretending certainty.

The AI diagnosis is decision support, not an unconditional prescription.

The scan result should open into a conversational thread so the farmer can continue asking questions without losing context.

10. Community

Community is a first-class product surface, not a comment section attached to another feature.

Public discussion threads

Example: “Should I water my crop this week?”

Other farmers can reply. Experts can also reply.

The UI must distinguish:

- Farmer response
- Verified/qualified expert response
- Dealer response, where relevant

The product must never visually imply that every answer has equal authority.

Stories / updates

Farmers can share lightweight updates similar to social stories/statuses.

Examples:

- Harvest update
- Crop progress
- New crop planted
- Farming activity

- Field photo
- Success/failure story
- Local conditions

Stories can support reactions and/or replies according to the final interaction model.

Community safety

Community advice can directly affect crop yield and money.

Therefore:

- Reporting is required.
- Moderation is required.
- Expert identity must be verifiable.
- Harmful/unsafe advice needs escalation.
- AI-generated answers must not masquerade as human expert advice.

11. Messages

Messages contain private communication, separate from public Community discussions.

Farmer ↔ Dealer

Use cases:

- Ask whether product is available.
- Ask current price.
- Ask about an order.
- Discuss credit request.
- Confirm pickup/delivery.
- Clarify product details.

This is a private thread.

Other private communication is future scope unless explicitly approved.

12. Dealer Discovery

When the farmer has a relevant need, show verified agricultural shops.

Ranking can consider:

- Distance
- Relevant stock
- Product availability
- Farmer's existing dealer relationship
- Shop verification status
- Operating status
- Delivery/pickup options

The platform should not imply that “nearby” means “recommended” unless the ranking logic supports that claim.

13. Product Discovery

Products can be surfaced from the existing dealer network.

Possible entry points:

- AI diagnosis
- Farmer search
- Dealer chat
- Shop/order flow
- Crop recommendations

Product information should make clear:

- Product name

- Relevant crop/problem where applicable
- Pack/variant
- Price
- Availability
- Dealer
- Verification/trust signals
- Pickup/delivery information
- Credit availability, if applicable

A product recommendation must not be interpreted as an endorsement merely because it appears in the app.

14. Credit — Critical Product System

Credit is not a hidden CTA, secondary modal, or informal chat agreement.

It is a formal, visible platform workflow between two parties.

Parties

- Farmer/requesting party
- Dealer/lending or credit-granting party
- Platform as workflow, record, visibility, and rules layer

The exact legal/financial role of the platform is not yet finalized and requires business/legal review before implementation.

Core principle

Neither party should rely on memory, verbal promises, or hidden system state.

Both parties must have clear visibility into the agreement.

Proposed credit lifecycle

Eligibility → Request → Terms → Review → Decision → Agreement → Fulfilment → Outstanding balance → Repayment → Closure

Every transition must have an explicit state.

Eligibility

The system determines whether the farmer can request credit under platform rules.

Potential inputs:

- Existing dealer relationship
- Previous purchase/payment history
- Existing outstanding amount
- Crop/season context
- Dealer-defined limits
- Platform risk rules
- Other approved eligibility factors

Internal risk scores should not automatically be exposed to the farmer.

Credit request

Farmer selects the relevant product/order and requests credit.

The request must clearly show:

- Requested amount
- Product/order
- Dealer
- Proposed repayment terms
- Any fees/markup
- Expected repayment event/date or condition
- Important conditions
- What happens if the request is rejected

Dealer/platform review

The request enters a visible pending state.

The dealer sees the information needed to make the decision.

The platform enforces required criteria before approval.

Decision

Possible states:

- Pending
- Approved
- Rejected
- More information required
- Expired
- Cancelled

A rejection should have a structured reason where appropriate.

Agreement

Approval is not the same as a completed agreement.

Before fulfilment, both parties see the final terms.

The farmer explicitly acknowledges/accepts.

The dealer confirms/accepts.

The system records:

- Who accepted
- When
- Exact terms
- Product/order
- Amount
- Repayment condition

- Applicable fees/markup
- Relevant policy/version

The agreement is stored as an immutable historical record. Amendments are new versions, not silent edits to the old agreement.

Fulfilment

Only after required agreement conditions are satisfied can the credit-backed order proceed.

Repayment

The system tracks:

- Original amount
- Amount paid
- Remaining amount
- Payment history
- Due condition/date
- Status

Closure

When outstanding reaches zero, the agreement becomes closed.

No workflow should allow paid amount to exceed valid outstanding.

Platform rules

The platform owns workflow integrity.

The dealer cannot bypass mandatory agreement states through an alternate UI/API path.

The farmer cannot create fake/duplicate credit agreements through retries or repeated requests.

Critical transitions must be enforced server-side, not only in UI.

Credit visibility

Farmer can always see:

- Active requests
- Approved agreements
- Current outstanding
- Payment history
- Agreement terms
- Status
- Next required action

Dealer can see the corresponding operational view.

The platform maintains the canonical record.

Credit terminology

Avoid ambiguous language such as “approved” when the legal/economic agreement has not yet been accepted by both parties.

Working state language:

Eligible → Requested → Approved → Agreement accepted → Fulfilled → Outstanding → Paid/Closed.

Exact terminology requires validation with the business/legal model.

15. Weather

Weather is a utility, not the product's identity.

The goal is to translate weather into useful farming decisions.

Examples:

- Spray suitability
- Rain risk
- Irrigation considerations

- Heat/cold alerts
- Wind conditions
- Crop-specific warnings

Avoid simply presenting raw weather numbers when an actionable interpretation is possible.

16. Daily Agricultural Updates

Content should be:

- Location-aware
- Crop-aware
- Season-aware
- Action-oriented
- Short
- Voice-friendly where appropriate

Avoid becoming a generic news/content feed.

17. Market / Crop Rates

Potential content:

- Local crop rates
- Relevant mandi/market information
- Price trends where reliable
- Timing context

Source and timestamp must be visible.

Never present stale or unverified rates as current facts.

18. Farmer / Farm Profile

Profile should eventually contain:

- Farmer identity
- Location
- Farms/fields
- Crops
- Crop stage
- Dealer relationships
- Orders
- Credit agreements
- Payment history
- Scan/diagnosis history
- Saved advice
- Notifications/settings

Farm profile is more important than a generic social profile because personalization depends on actual agricultural context.

19. Crop / Farm Context Model

The product should progressively build useful context:

Farmer → Farm → Field → Crop → Variety → Season → Growth stage → Activities → Problems → Inputs → Outcomes

Do not force farmers to enter all of this during onboarding. Progressive profiling is preferred.

20. Notifications

Notifications should be event-driven and actionable.

Examples:

- Weather warning

- Spray window
- Crop activity reminder
- AI diagnosis result
- Expert response
- Community response
- Dealer reply
- Credit request status
- Agreement requiring action
- Payment/repayment update
- Order status

Do not create notification spam.

21. Accessibility / Real-World Constraints

Design for:

- Outdoor glare
- One-handed use
- Large touch targets
- Low reading comfort
- Urdu and regional languages
- Voice/audio
- Shared devices
- Weak/intermittent connectivity
- Low-cost Android devices

Offline-first behavior should be considered for critical capture and draft workflows.

22. Trust & Safety

Trust is a system property, not a badge alone.

Trust signals can include:

- Verified dealer
- Verified expert
- Product/source information
- Timestamp
- Location relevance
- Community reputation where validated
- Clear uncertainty from AI
- Clear credit terms
- Visible transaction history

Never use a badge as decoration without a defined verification process behind it.

23. AI Rules

AI must:

- Separate known facts from assumptions.
- Communicate uncertainty.
- Ask for missing high-value context rather than guessing.
- Never invent product availability, pricing, expert status, weather, or credit eligibility.
- Use canonical backend data for transactional information.
- Never silently change business rules.
- Never make a financial commitment on behalf of either party.
- Never approve credit unless the defined authorization system explicitly permits it.

- Preserve conversation context where needed.
- Escalate low-confidence/high-risk cases.
- Avoid presenting AI output as verified expert advice.

24. Design Thinking Operating System

Every feature, screen, and component follows this sequence.

Why

Why does this exist? What user/business problem does it solve? What happens if we remove it?

User intent

What is the user trying to accomplish right now? What is the user's mental model? What information is needed to make the decision?

Context

What is happening before this screen? What happens after it? What data/state does the system know? What does it not know?

Information priority

Rank information:

1. Critical
2. Action-enabling
3. Important context
4. Supporting detail
5. Optional

Never treat every data point equally.

Interaction model

Define:

- Primary action
- Secondary action
- Escape/cancel
- Error
- Empty
- Loading
- Success
- Failure
- Retry
- Permission/access
- Offline behavior

System logic

For every visible state ask:

- Where does this data come from?
- Who owns it?
- Is it calculated or stored?
- When is it updated?
- What is the source of truth?
- What happens if it is missing/stale?
- Can the user manipulate it?
- What validation is required?

Second-order effects

Before finalizing a decision, ask:

- What does this enable next?
- What can go wrong because of this?
- Does it create a new user expectation?
- Does it create operational work?
- Does it create fraud/abuse risk?
- Does it affect trust?
- Does it affect data consistency?
- Does it create legal/financial implications?
- Does it duplicate an existing system?
- What happens at scale?

Trade-off

Record:

- Options considered
- Chosen direction
- What we sacrifice
- Conditions under which we would revisit it

25. Component-Level Design Rules

A component is not complete because it looks good.

For each reusable component define:

- Purpose
- Data required
- States
- Variants
- Interaction behavior

- Responsive behavior
- Accessibility
- Loading/error/empty behavior
- RTL behavior
- Localization considerations
- Analytics/event needs where relevant
- Backend dependency
- Security/trust implications

The design system should remain centralized and token-based.

One change to a shared token/component should propagate across usages rather than being manually recreated screen by screen.

26. Source-of-Truth Rules

Every important field must have one canonical source.

Examples:

- Product availability → inventory system
- Product price → defined pricing source
- Credit balance → credit ledger
- Agreement terms → agreement record/version
- Dealer verification → dealer verification system
- Expert verification → expert verification system
- Weather → defined weather provider
- Market rate → defined market source
- AI diagnosis → AI diagnosis service + stored case context

The UI must not invent or independently maintain authoritative business data.

27. Backend / CTO Review Gate

Before implementation of any non-trivial feature, Product + Design + Engineering/CTO review:

1. User problem
2. Business rule
3. Data model impact
4. Source of truth
5. Permissions
6. State machine
7. Failure modes
8. Security
9. Audit/history
10. Retry/idempotency
11. Migration/backward compatibility
12. Operational impact
13. Performance
14. Cost
15. Third-party dependency

For high-risk workflows such as credit, payments, identity, or product recommendations, add legal/compliance/business review where required.

28. State-Machine Rule

Any feature with meaningful lifecycle states must explicitly define its state machine.

Example:

Credit Request → Pending → Approved → Agreement Pending → Accepted → Fulfilled → Outstanding → Partially Paid → Closed

Invalid transitions must be blocked server-side.

Examples:

- Closed agreement cannot be edited as active.
- Rejected request cannot silently become fulfilled.
- Duplicate submission cannot create duplicate agreement.
- Paid amount cannot exceed outstanding.
- Agreement terms cannot be silently changed after acceptance.
- Cancelled request cannot be fulfilled.

29. Decision Log

Every important product/system decision should be recorded with:

Decision ID

Date

Decision

Context at the time

Problem

Options considered

Chosen approach

Why

Trade-offs

Dependencies

What would cause us to revisit it

Status: Proposed / Accepted / Superseded

When a decision changes, do not simply overwrite history.

Create a new decision referencing the previous decision and explain:

- What changed

- Why it changed
- What new evidence/context caused the change
- What existing work is affected

30. Scope Control

Every new feature/request must answer:

1. What user problem does it solve?
2. Which product goal does it support?
3. Which existing journey does it belong to?
4. What is the smallest useful version?
5. What dependencies does it create?
6. What does it replace or delay?
7. What operational burden does it add?
8. What data does it require?
9. What risks does it introduce?

If it cannot answer these questions, it remains an idea rather than becoming committed scope.

31. MVP Working Scope

The initial core product should prove:

Crop problem → Scan/understand → Guidance → Relevant product/dealer → Optional credit/order → Fulfilment → Record/history

Supporting capabilities required for a meaningful experience:

- Home
- Scan/AI
- Dealer discovery

- Product discovery
- Dealer chat
- Credit workflow
- Basic farmer/farm context
- Notifications
- Community foundation

Weather, daily content, market rates, stories, and richer community functionality can be phased according to validation and implementation capacity.

This is a prioritization hypothesis, not a final launch commitment.

32. Out of Scope Until Explicitly Approved

Do not automatically add:

- Full open marketplace
- Unverified sellers
- Independent credit scoring product
- Complex logistics network
- Bank replacement
- Insurance product
- Full social network
- Full agricultural ERP
- Original computer-vision research/model training
- Unbounded AI advisory

33. Current Open Decisions

Owner decisions

None remaining for the current MVP product definition.

The following are locked and should not be repeatedly reopened:

- Farmer-first / guest-first access with identity requested at the point an accountable action requires it.
- Farmer ↔ Shop Owner credit workflow, including Shop Owner-controlled terms, farmer consent to terms, one counteroffer maximum, farmer accept/reject, and Shop Owner final approval.
- Privacy/AI/media consent behavior where consent is required by the purpose of collection or use.
- Verified Expert requires verified identity + at least one relevant, verifiable professional qualification/certification.
- Admin/Moderation is the final authority for expert verification and community moderation.
- Current scope, geography, languages, dealer radius, source-of-truth rules, recommendation architecture and flow-simplification/change-control rules.

External / operational dependencies

These are not product-discovery questions. Architecture/engineering handles them and escalates only when an actual account purchase, contract, legal review or external approval is required:

- final legal wording and formal privacy policy;
- provider account/API credentials and permitted commercial access;
- AMIS data-access method/permission;
- final analytics/offline implementation details;
- future scaling of credit/financial products;
- certificate expiry/re-verification operational policy.

Architecture-owned proposals already defined

- Agricultural knowledge coverage tiers + initial priority crop roadmap.
- Weather provider abstraction with **WeatherAPI.com Free as the initial MVP candidate** (100K calls/month, 3-day forecast, current weather, commercial

use allowed under its free-plan terms). Upgrade only if scale/features require it. Open-Meteo remains a free non-commercial evaluation/prototyping option, not the assumed commercial production free tier. cite turn215206search1 turn215206search10 turn215206search2

- Market adapter using AMIS where permitted machine-readable access is available.
- Text support for Urdu/Punjabi/Saraiki/Sindhi/Pashto; voice capability only where provider quality is validated, with Saraiki voice requiring additional validation.
- Read-heavy offline support rather than full offline transaction operation.
- PostHog as analytics candidate with sensitive AI/credit payloads excluded.
- Expert certificate expiry/re-verification data model and moderation escalation mechanics.

These do not block the PRD from being implementation-ready; they are tracked as research/dependency items and must not be silently converted into legal policy.

34. Current Design Evidence

Design screens already exist and will be added here as the visual source of truth.

Planned evidence:

- Home
- Scan / AI chat
- AI diagnosis result
- Dealer/product match
- Dealer chat
- Credit request
- Credit agreement
- Credit status/history
- Community

- Community thread
- Stories/status
- Profile
- Orders
- Notifications

When design evidence is added, decisions should be checked against the actual UI rather than relying only on this textual PRD.

35. AI Agent / Implementation Handoff Model

Any AI agent working on this product must treat this page as a product source of truth, but must not assume every section is finalized.

The agent must distinguish:

Known / Accepted — explicitly decided.

Proposed — recommended but not approved.

Open Decision — requires discussion.

Evidence Required — cannot be decided safely without validation.

Superseded — historical decision no longer active.

The agent must never silently convert a proposal into a requirement.

Before implementing a feature, inspect:

1. Product requirements
2. Current decisions
3. Current design evidence
4. Relevant backend/business rules
5. Existing implementation
6. Dependencies

7. Existing data/source-of-truth rules

If something conflicts with an earlier accepted decision, stop and surface the conflict.

36. Required Agent Behaviour

The agent acts as two connected roles.

Product Designer

Responsible for:

- User intent
- Journey
- Information hierarchy
- Interaction design
- Accessibility
- Design-system consistency
- UX trade-offs
- Validation questions

Technical/Product Leader

Responsible for:

- Architecture implications
- Data ownership
- State transitions
- Security
- Permissions
- Consistency
- Failure modes

- Scalability
- Cost
- Maintainability
- Implementation feasibility

The agent should challenge both the user's assumptions and its own assumptions.

It should not agree simply because an idea sounds reasonable.

37. Working Process From Here

When a new feature or implementation requirement is introduced:

1. Extract intent.
2. Identify the user/business problem.
3. Check existing decisions.
4. Check existing design.
5. Identify dependencies.
6. Identify unknowns.
7. Apply second-order thinking.
8. Propose the smallest coherent solution.
9. Identify risks/trade-offs.
10. Decide what needs user/business/technical approval.
11. Update the PRD/decision log after the decision is actually made.
12. Keep implementation scope aligned with the approved decision.

When asked “where does this data come from?”, answer from the system's defined source of truth. If no source has been defined, say so and create an explicit open decision rather than inventing one.

38. Definition of Done

A feature is not done when the happy-path UI exists.

It is done when:

- User goal is clear.
- Business rule is explicit.
- Design covers required states.
- Backend source of truth is defined.
- Permissions are defined.
- Invalid transitions are blocked.
- Error/retry behavior is defined.
- Loading/empty/offline states are defined where relevant.
- Audit/history requirements are defined where relevant.
- Analytics requirements are defined where relevant.
- Design system usage is consistent.
- Localization/RTL implications are considered.
- Implementation is validated against the intended journey.
- Important assumptions have evidence or are explicitly marked as assumptions.

39. Working Decision Record — Credit

Status: Proposed → requires business/legal confirmation.

Current direction: Credit requests are a formal, visible workflow. The farmer cannot silently request or receive credit without the system clearly representing request, review, decision, agreement, fulfilment, outstanding amount, repayment, and closure.

Key principle: Both farmer and dealer must have visibility into the exact terms and status.

Platform role: The platform should enforce the workflow and record the agreement. Whether the platform is merely facilitating dealer credit or directly

participating in credit provision remains an open business/legal decision.

Reason: Credit affects money, trust, disputes, and potentially legal obligations. It cannot be treated as a normal marketplace interaction.

40. Product North Star

The product should eventually feel like:

“My farming companion knows what I grow, understands what is happening, helps me decide what to do, connects me to people I trust, and keeps my important farming decisions and transactions clear.”

Not:

“Another agriculture marketplace with an AI chatbot.”

41. Immediate Next Work

1. Add the existing designed screens as visual evidence.
2. Map each screen to the user journey.
3. Identify gaps between current design and this PRD.
4. Finalize the MVP journey.
5. Define the first pilot crop/location.
6. Resolve the credit business/legal model before implementation of real credit.
7. Define the initial backend domain model and source-of-truth boundaries.
8. Create AI-agent operating files from this source of truth:
    - ROLE.md
    - PRODUCT_RULES.md
    - DESIGN_THINKING.md
    - DECISIONS.md
    - ARCHITECTURE.md
    - IMPLEMENTATION_RULES.md

- QA_RULES.md

9. Keep those agent files synchronized with accepted decisions from this document.

42. Important Working Rule

Do not let implementation pressure silently make product decisions.

If implementation requires a decision that has not been made, surface it.

If a technical constraint forces a product trade-off, surface it.

If a new requirement conflicts with an earlier decision, surface both the old decision and the new context before changing it.

The goal is not merely to ship screens.

The goal is to build a coherent product whose user experience, business rules, data model, technical architecture, and operational reality agree with each other.

🧠 🧠 Engineering AI Operating System — Roles, Architecture & Technology Baseline

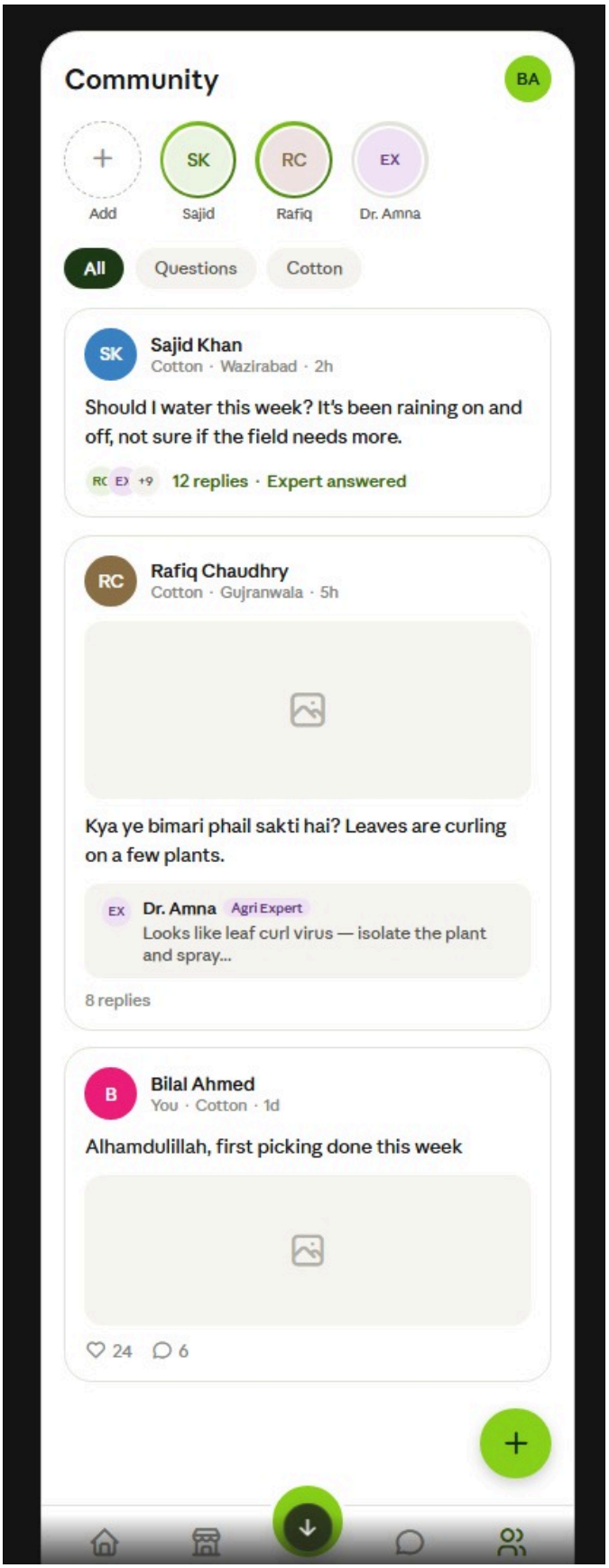
43. Engineering AI Operating System

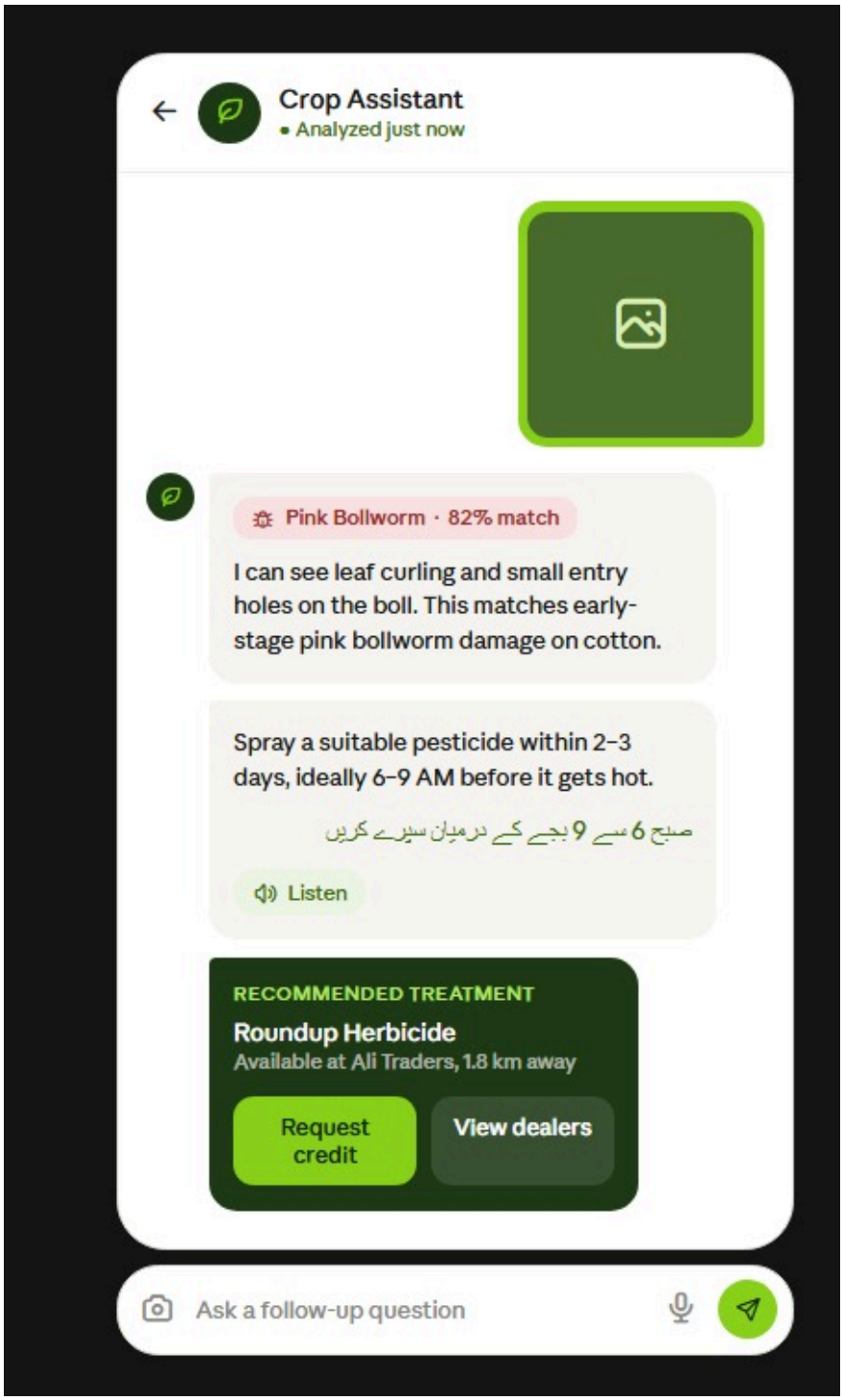
The persistent engineering/AI roles, technology baseline, QA architecture, autonomy tiers, evidence rules, engineering memory model, and reconstructed source-document knowledge are maintained in the child page:

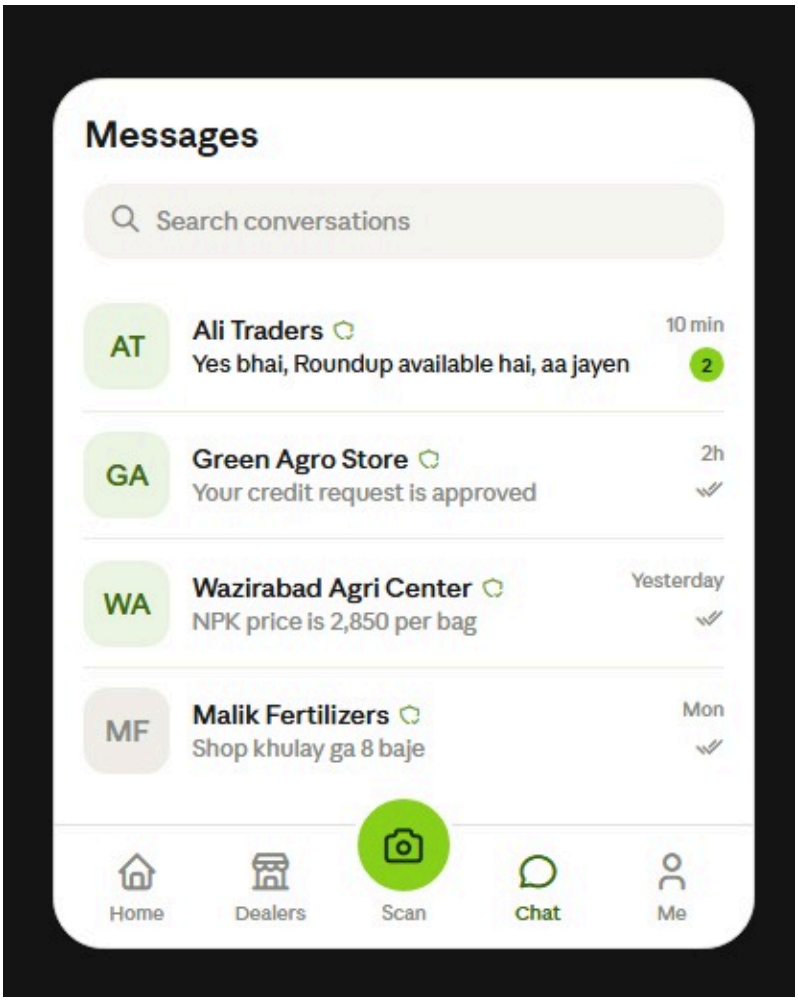
Engineering AI Operating System — Roles, Architecture & Technology Baseline

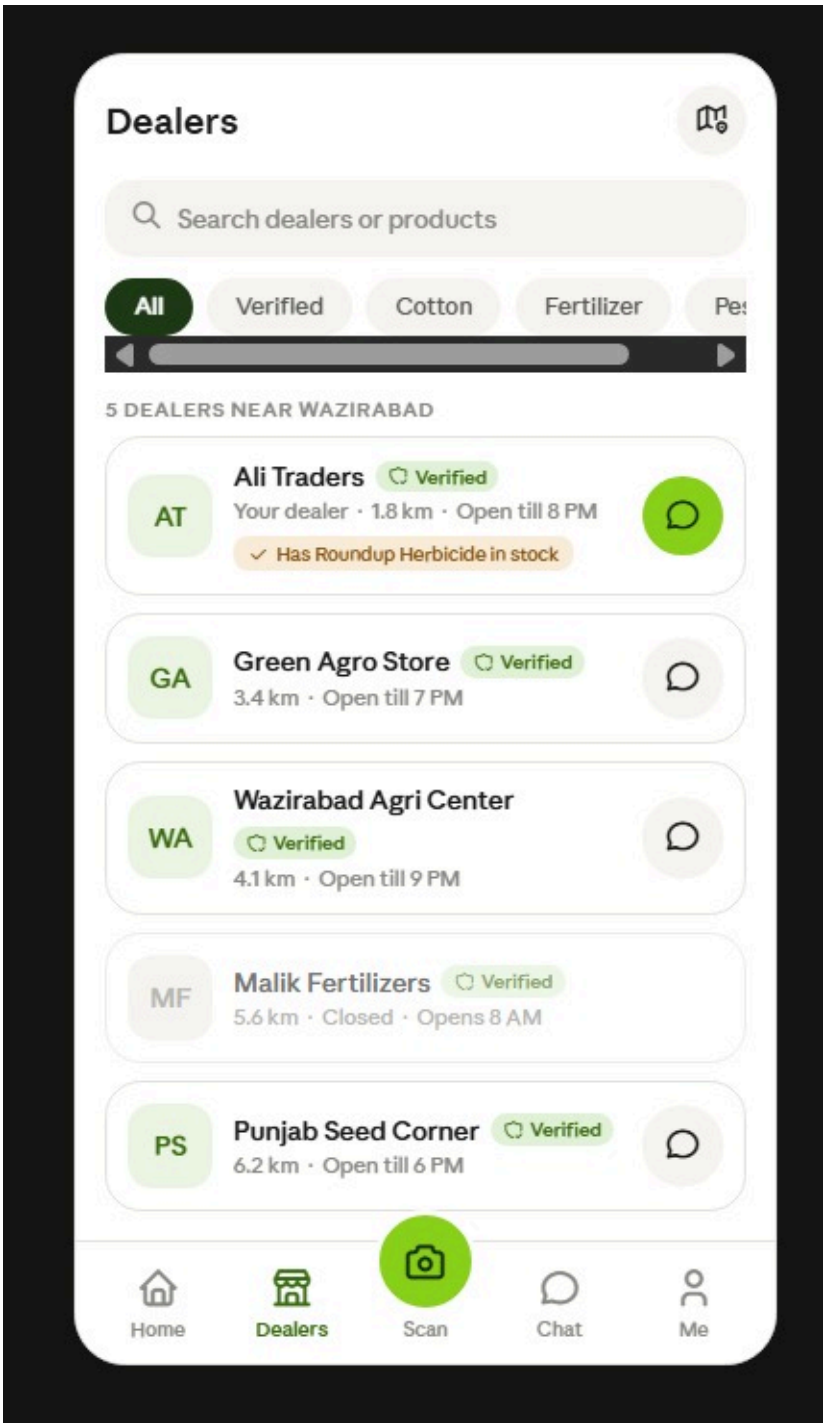
This page is the operating reference for future AI implementation and QA work. It distinguishes locked decisions from current baselines, proposals, open decisions, evidence-required items, and superseded history.

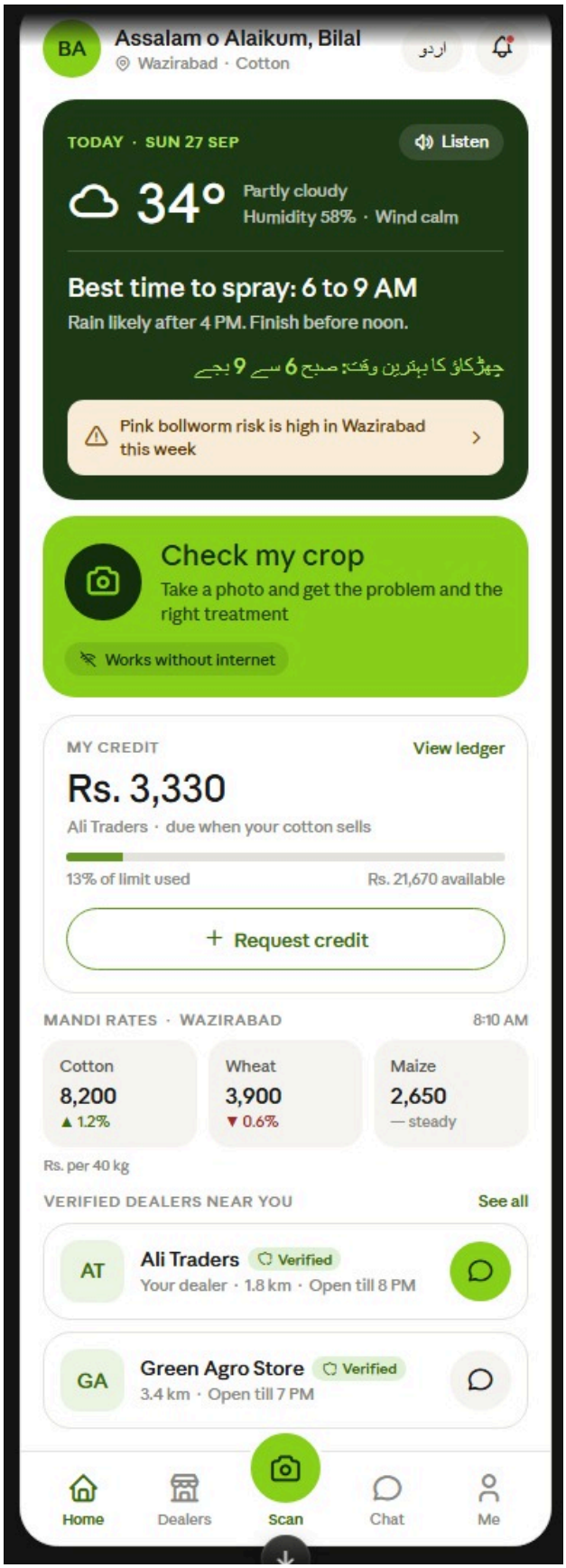
43. App final design screens

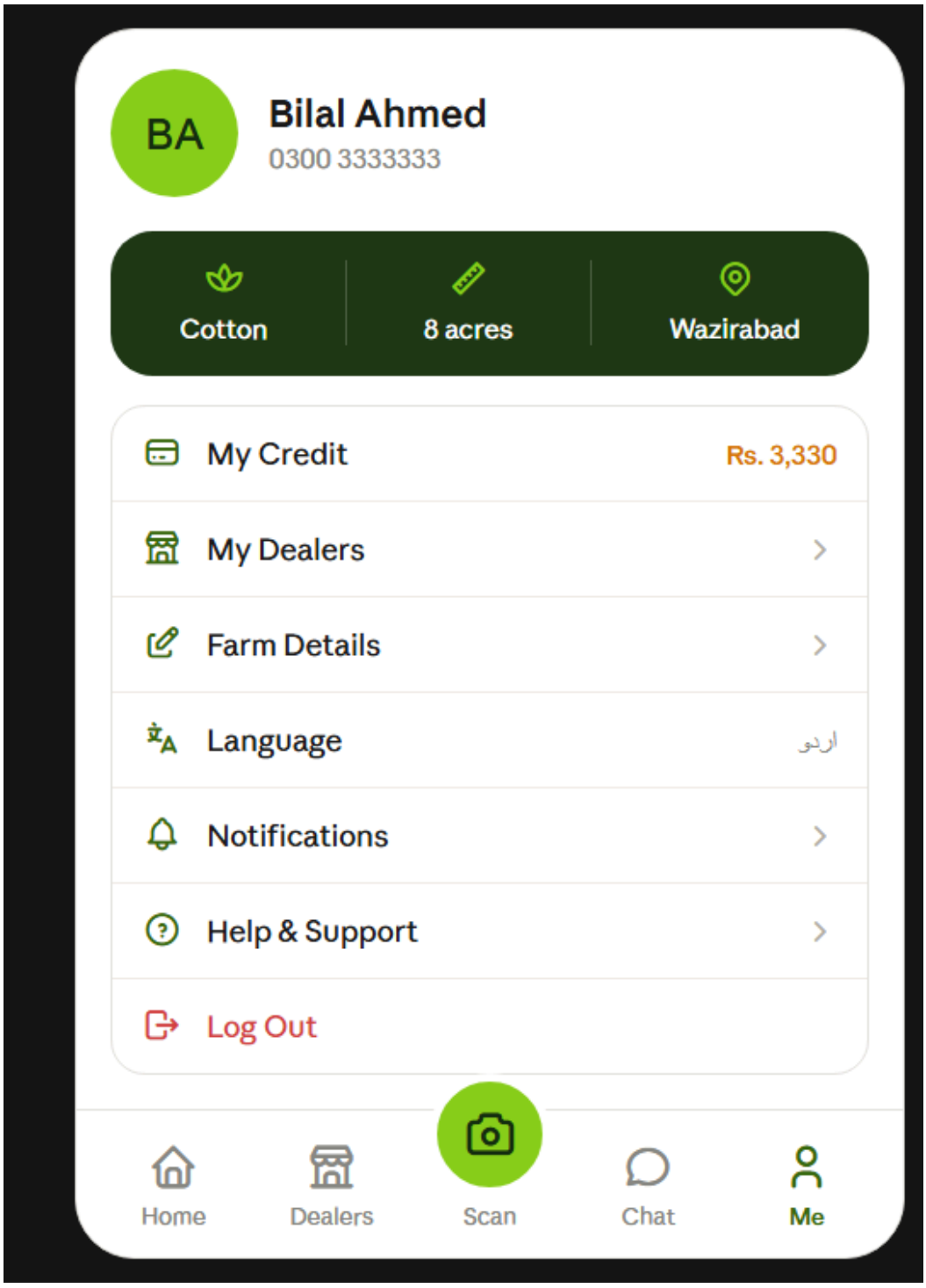

Community
BA
Add
SK
Sajid
RC
Rafiq
EX
Dr. Amna
All
Questions
Cotton
SK
Sajid Khan
Cotton · Wazirabad · 2h
Should I water this week? It's been raining on and off, not sure if the field needs more.
RC EX +9
12 replies · Expert answered
RC
Rafiq Chaudhry
Cotton · Gujranwala · 5h
Kya ye bimari phail sakti hai? Leaves are curling on a few plants.
EX
Dr. Amna
Agri Expert
Looks like leaf curl virus — isolate the plant and spray...
8 replies
B
Bilal Ahmed
You · Cotton · 1d
Alhamdulillah, first picking done this week
24
6


Crop Assistant
• Analyzed just now
Pink Bollworm · 82% match
I can see leaf curling and small entry holes on the boll. This matches early-stage pink bollworm damage on cotton.
Spray a suitable pesticide within 2–3 days, ideally 6–9 AM before it gets hot.
صبح 6 سے 9 بجے کے درمیان سپرے کریں
Listen
RECOMMENDED TREATMENT
Roundup Herbicide
Available at Ali Traders, 1.8 km away
Request credit
View dealers
Ask a follow-up question


Messages
Search conversations
AT
Ali Traders
Yes bhai, Roundup available hai, aa jayen
10 min
2
GA
Green Agro Store
Your credit request is approved
2h
WA
Wazirabad Agri Center
NPK price is 2,850 per bag
Yesterday
MF
Malik Fertilizers
Shop khulay ga 8 baje
Mon
Home
Dealers
Scan
Chat
Me


Dealers
Search dealers or products
All
Verified
Cotton
Fertilizer
5 DEALERS NEAR WAZIRABAD
AT
Ali Traders
Verified
Your dealer · 1.8 km · Open till 8 PM
Has Roundup Herbicide in stock
GA
Green Agro Store
Verified
3.4 km · Open till 7 PM
WA
Wazirabad Agri Center
Verified
4.1 km · Open till 9 PM
MF
Malik Fertilizers
Verified
5.6 km · Closed · Opens 8 AM
PS
Punjab Seed Corner
Verified
6.2 km · Open till 6 PM
Home
Dealers
Scan
Chat
Me


BA
Assalam o Alaikum, Bilal
Wazirabad · Cotton
اردو
TODAY · SUN 27 SEP
Listen
34°
Partly cloudy
Humidity 58% · Wind calm
Best time to spray: 6 to 9 AM
Rain likely after 4 PM. Finish before noon.
چھڑکاؤ کا بہترین وقت: صبح 6 سے 9 بجے
Pink bollworm risk is high in Wazirabad this week
Check my crop
Take a photo and get the problem and the right treatment
Works without internet
MY CREDIT
View ledger
Rs. 3,330
Ali Traders · due when your cotton sells
13% of limit used
Rs. 21,670 available
+ Request credit
MANDI RATES · WAZIRABAD
8:10 AM
Cotton
8,200
▲ 1.2%
Wheat
3,900
▼ 0.6%
Maize
2,650
— steady
Rs. per 40 kg
VERIFIED DEALERS NEAR YOU
See all
AT
Ali Traders
Verified
Your dealer · 1.8 km · Open till 8 PM
GA
Green Agro Store
Verified
3.4 km · Open till 7 PM
Home
Dealers
Scan
Chat
Me


BA
Bilal Ahmed
0300 3333333
Cotton
8 acres
Wazirabad
My Credit
Rs. 3,330
My Dealers
Farm Details
Language
اردو
Notifications
Help & Support
Log Out
Home
Dealers
Scan
Chat
Me